

FedICL-MQA: A Federated In-Context Learning Framework using Small Language Models for Medical Question Answering

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Abstract. A concise (150–250 words) summary.

- Background and motivation
- Challenges in Medical Question Answering (MQA)
- Limitations of centralized medical QA systems
- Motivation for federated in-context learning using Small Language Models (SLMs)
- Overview of the proposed FedICL-MQA framework
- Experimental results and implications

1. INTRODUCTION

- Background
 - Evolution of Medical Question Answering systems
 - Role of NLP and language models in healthcare
 - Importance of trustworthy clinical QA systems
- Challenges in Existing Medical QA Systems
 - Patient privacy and data governance
 - Limited access to high-quality medical annotations
 - Computational costs of Large Language Models
 - Data heterogeneity across healthcare institutions
 - Domain adaptation challenges
- Motivation
 - Advantages of federated learning in healthcare
 - Emergence of in-context learning (ICL)
 - Potential of Small Language Models for clinical deployment
 - Need for scalable and privacy-preserving MQA systems
- Research Questions
 - RQ1: How can federated learning and in-context learning be effectively integrated to enable privacy-preserving Medical Question Answering across distributed healthcare institutions?
 - RQ2: To what extent can Small Language Models leverage in-context demonstrations to achieve competitive medical reasoning performance?
 - RQ3: How robust is the proposed framework under heterogeneous and non-IID medical data distributions?

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- Research Objectives
 - Develop a federated MQA framework without sharing raw patient data.
 - Enhance medical reasoning through in-context learning.
 - Improve SLM performance using adaptive demonstrations.
 - Achieve robustness under heterogeneous healthcare environments.
- Main Contributions
 - A novel Federated In-Context Learning framework for Medical Question Answering.
 - An adaptive medical demonstration retrieval mechanism.
 - A federated SLM-based learning architecture for privacy-preserving MQA.
 - A context-aware knowledge-sharing strategy across institutions.
 - Comprehensive evaluations on benchmark medical QA datasets.

2. RELATED WORK

2.1 Medical Question Answering

2.2 Federated Learning in Healthcare NLP

2.3 In-Context Learning

2.4 Small Language Models

2.5 Medical Knowledge Retrieval

2.6 Research Gap Analysis

3. PROPOSED APPROACH

3.1 Problem Formulation

3.2 Framework Overview

3.3 Multimodal Representation Learning

3.4 Adaptive Demonstration Retrieval

3.5 Federated In-Context Learning Mechanism

3.6 Federated Knowledge Sharing

3.7 Training Objective

4. EXPERIMENTS

4.1 Experimental Setup

- Datasets:

- MedQA
- PubMedQA

- SLMs:

- TinyLlama
- Phi-3 Mini
- Gemma-2B
- Qwen-2.5 Small
- ...
- Main baselines:
 - Centralized SLM
 - Federated SLM without ICL
 - LLM prompting baseline
 - Non-federated ICL
- Evaluation Metrics:
 - Answer Generation Metrics:
 - Accuracy,
 - F1-score,
 - BLEU,
 - ROUGE-L,
 - BERTScore
 - Federated Learning Metrics:
 - Communication Cost,
 - Convergence Rate
 - Efficiency Metrics:
 - Inference Latency,
 - GPU Memory Usage,
 - FLOPs
- Implementation details

4.2 Results

- Overall Performance Comparison (with baseline methods)
- Effectiveness of In-Context Learning
 - Impact of Demonstration Number
 - Impact of Context Length
 - Impact of Retrieval Strategy
- Federated Learning Analysis
- Small Language Model Analysis

- Comparison among:
 - TinyLlama
 - Phi-3 Mini
 - Gemma-2B
 - Qwen-2.5 Small
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- Ablation Studies

- FedICL-MQA w/o ICL
- FedICL-MQA w/o Federated Learning
- Different Retrieval Sizes
- Different Aggregation Strategies

4.3 Discussion

- Clinical Implications

- Advantages of the Proposed Framework

- Limitations

5. CONCLUSION

- Summary of the FedICL-MQA framework
- Key findings
- Contributions to federated healthcare NLP
- Future Directions
 - Federated retrieval-augmented generation
 - Continual medical learning
 - Agent-based medical QA
 - Personalized healthcare assistants
